

**DEPARTMENT OF INDUSTRIAL & MANAGEMENT ENGINEERING**

**DMS201: INTRODUCTION TO OPERATIONS MANAGEMENT**

**QUIZ # 02 |2025-2026, SEMESTER III|**

**NOTE THE FOLLOWING**

- 01) Total number of questions is **THIRTY (30)**.
- 02) Questions are: True/False; Range Based (Qualitative & Quantitative), MCQ (FIVE (05) options with ONE (01) or MORE CORRECT answer(s)).
- 03) Marks: CORRECT is +2/Plus TWO; WRONG is –1/Minus ONE; No ANSWER is 0
- 04) Time duration is FORTY FIVE (45) / SIXTY (40/54) minutes for non-DAP / DAP students.
- 05) Closed Notes/Book examination.

For **ABC Analysis of Inventory**, the graph of cumulative distribution of % inventory value (**depicted along Y-axis**) against product category (**depicted along X-Axis**) closely resembles the CDF of **Log-Normal Distribution**

True

You are given product-wise **Cost per Unit** (INR/Unit) and **Annual Usage** (Units). From this information find out which **product(s)** can be classified as category *A* item(s) under **ABC Analysis of Inventory**

| Product | Cost/Item (INR/Unit) | Annual Usage (Units) |
|---------|----------------------|----------------------|
| P1      | 5                    | 1000                 |
| P2      | 15                   | 200                  |
| P3      | 20                   | 1000                 |
| P4      | 5                    | 4500                 |
| P5      | 10                   | 750                  |
| P6      | 25                   | 500                  |

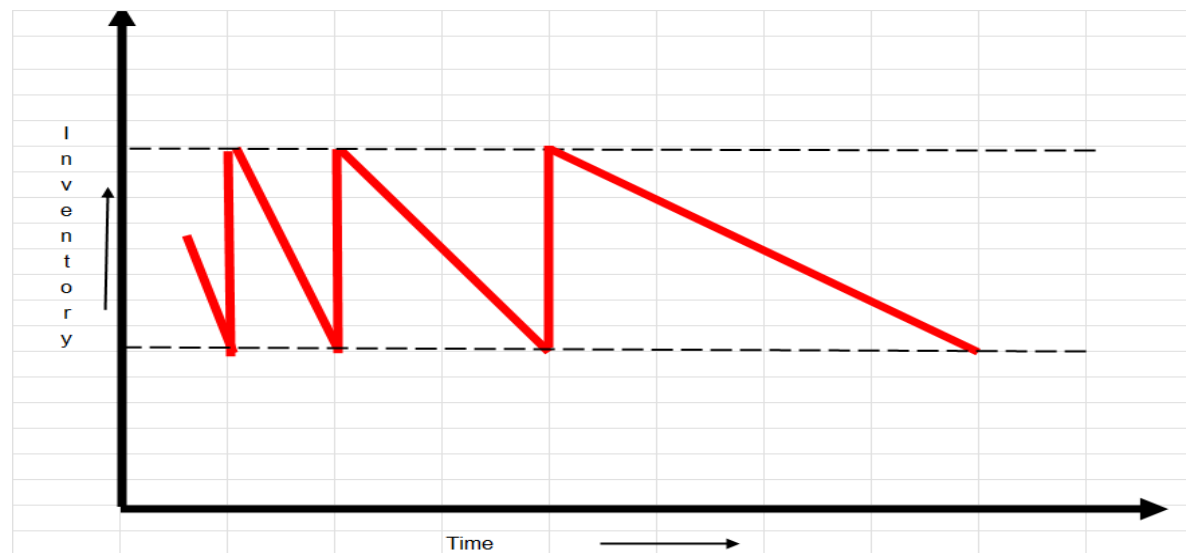
|   |                    |
|---|--------------------|
| A | P4, P3, P6         |
| B | P3, P6, P5, P1     |
| C | P4, P6, P5, P2     |
| D | P3, P5, P1         |
| E | P4, P3, P1, P2, P5 |

| Product | Cost/Item (INR/Unit) | Annual Usage (Units) | Annual Vale (INR) |
|---------|----------------------|----------------------|-------------------|
| P1      | 5                    | 1000                 | 5000              |
| P2      | 15                   | 200                  | 3000              |
| P3      | 20                   | 1000                 | 20000             |
| P4      | 5                    | 4500                 | 22500             |
| P5      | 10                   | 750                  | 7500              |
| P6      | 25                   | 500                  | 12500             |

| Product | Cost/Item (INR/Unit) | Annual Usage (Units) | Annual Vale (INR) | % of Total | Cumulative% |
|---------|----------------------|----------------------|-------------------|------------|-------------|
| P4      | 5                    | 4500                 | 22500             | 31.91      | 31.91       |
| P3      | 20                   | 1000                 | 20000             | 28.37      | 60.28       |
| P6      | 25                   | 500                  | 12500             | 17.73      | 78.01       |
| P5      | 10                   | 750                  | 7500              | 10.64      | 88.65       |
| P1      | 5                    | 1000                 | 5000              | 7.09       | 95.74       |
| P2      | 15                   | 200                  | 3000              | 4.26       | 100.00      |
|         |                      |                      | 70500             | 100        |             |

For the following graph (Inventory Level (**depicted along Y-axis**) versus Time (**depicted along X-axis**)) which of the following statement is TRUE.

- **Statement I:** (i) Instantaneous order rate; (ii) Constant usage rate with respect to time; (iii) Zero Level of Safety Inventory whenever an order is made
- **Statement II:** (i) Instantaneous order rate; (ii) Decreasing usage rate with respect to time; (iii) Positive Level of Safety Inventory whenever an order is made
- **Statement III:** (i) Instantaneous order rate; (ii) Constant usage rate with respect to time; (iii) Positive Level of Safety Inventory whenever an order is made
- **Statement IV:** (i) Instantaneous order rate; (ii) Instantaneous usage rate with respect to time; (iii) Zero Level of Safety Inventory whenever an order is made
- **Statement V:** (i) Instantaneous order rate; (ii) Decreasing usage rate with respect to time; (iii) Zero Level of Safety Inventory whenever an order is made

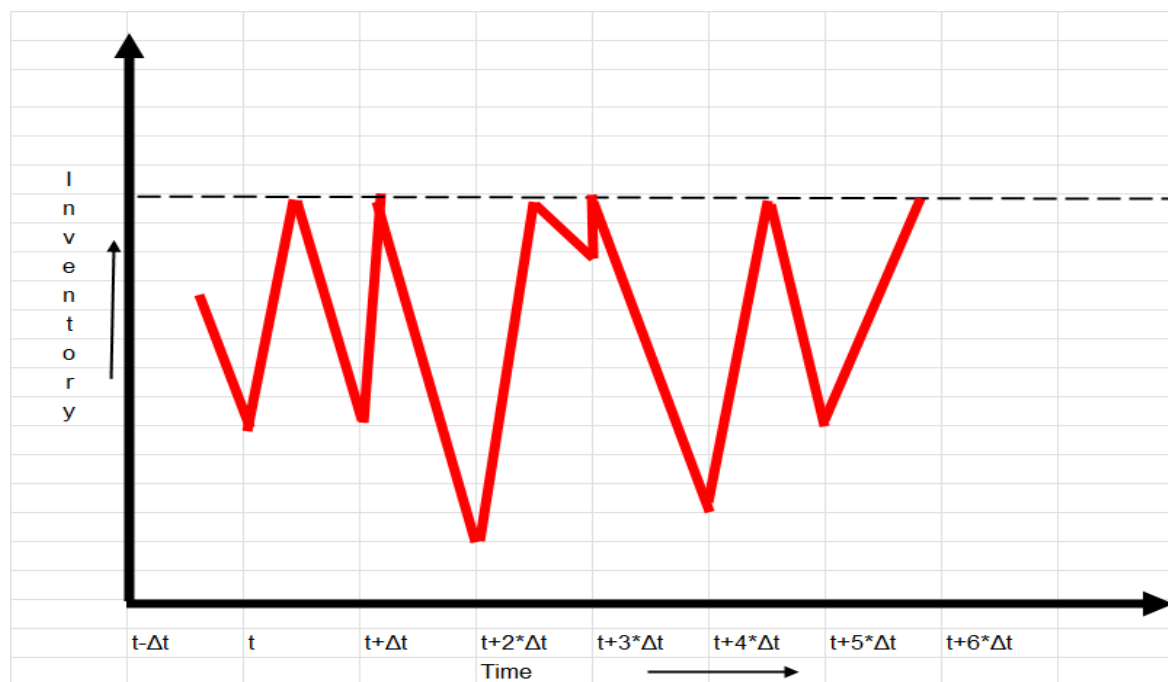


|   |               |
|---|---------------|
| A | Statement I   |
| B | Statement II  |
| C | Statement III |
| D | Statement IV  |
| E | Statement V   |

## Statement II

For the following graph (Inventory Level (depicted along Y-axis) versus Time (depicted along X-axis) which of the following statement is **TRUE**.

- **Statement I:** (i) Instantaneous order rate; (ii) Varying usage rate with respect to time; (iii); Time between two consecutive orders is varying
- **Statement II:** (i) Instantaneous order rate; (ii) Varying usage rate with respect to time; Time between two consecutive orders is fixed
- **Statement III:** (i) Varying order rate; (ii) Varying usage rate with respect to time; (iii) Time between two consecutive orders is fixed
- **Statement IV:** (i) Varying order rate; (ii) Varying usage rate with respect to time; (iii) Time between two consecutive orders is varying
- **Statement V:** (i) Varying order rate; (ii) Decreasing usage rate with respect to time; (iii) Time between two consecutive orders is fixed



|   |               |
|---|---------------|
| A | Statement I   |
| B | Statement II  |
| C | Statement III |
| D | Statement IV  |
| E | Statement V   |

## Statement III

**Anticipation** inventory/stock results when items compete for limited capacity of men, machines and

resources, plus also when significant setup times are there

False

To produce 37000 units of a particular 100 hp motor the factory is **designed** so that it operates 24 days a month working on two shifts, each of eight hours per day. It was later observed that average **effective** capacity is 95 units (of the same motor) per working hour. Furthermore, due to quality problems and related issues the percentage rejection on average is found to be 5%. Based on the information provided, find **utilization rate** (in percentage terms and to two places of decimal)

Range 93.65 to 93.69

Design output/capacity = 37000

Effective output/capacity =  $95 \times 8 \times 2 \times 24 = 36480$

Actual output/capacity =  $0.95 \times 36480 = 34656$

Hence utilization =  $(\text{Actual output} / \text{Design capacity}) \times 100\% = (34656 / 37000) \times 100 = 93.67\%$

The information on the current design of an aluminum plant shows that it should produce 12000 tonnes of aluminum ingots per month if the plant is open 30 days a month (each month has 30 days) working three shifts a day (each shift is eight hours each). Due to supply and other constraints, it is observed that **effective** capacity is 95% of the **design** capacity. The actual output observed for a 1 year period is 135000 tonnes. Find the **efficiency** of the above aluminum plant (in percentage terms and to two places of decimal)

Range 98.48 to 98.88

Design capacity =  $12000 \times 12 = 144000$

Effective capacity =  $0.95 \times 144000 = 136800$

Actual capacity/output = 135000

Efficiency =  $(\text{Actual output} / \text{Effective capacity}) \times 100\% = (135000 / 136800) \times 100 = 98.68\%$

Given fixed cost as Rs. 20,000/-, revenue per unit as Rs. 5/- and variable cost per unit as Rs. 3/-. What is the break even point?

Range 9999.99 to 10000.01

$Q_{BEP} = FC / (r - v)$ , when  $P = 0$ , hence  $Q_{BEP} = 20000 / (5 - 3) = 20000 / 2 = 10,000$

Arvind Mills which manufacture raw material for jeans wants to find the business strategy for selling specialized new raw material, for which the company requires setting up a new plant which would be at a fixed cost of Rs. 150 Crores. Variable costs would be Rs. 22,500/- per 100 meters of raw materials produced which would sell for Rs. 275/- per meter. In case the production is 31,000,000

meters then what is the total profit (in Rs. Crores), where total profit is defined as total revenue minus total cost?

Range 4.99 to 5.01

Break Even Point (BEP) =  $FC/(r - v) = 150 \times 10,000,000 / (275 - 225) = 30,000,000$  meters

If production is 31,000,000 then profit is  $(31,000,000 - 30,000,000) \times (275 - 225) = 10,000,000 \times 50 = 500,000,000 = 5$  Crores

Arvind Mills which manufacture raw material for jeans wants to find the business strategy for selling specialized new raw material, for which the company requires setting up a new plant which would be a fixed cost of Rs. 150 Crores. Variable costs would be Rs. 22,500/- per 100 meters of raw materials produced which would sell for Rs. 275/- per meter. In case the company wishes to earn a **total profit** of Rs. 6 Crores, then what would be the **total cost** (in Rs. Crores), where **total profit** is defined as **total revenue** minus **total cost**?

Range 701.99 to 702.01

Break Even Point (BEP) =  $FC/(r - v) = 150 \times 10,000,000 / (275 - 225) = 30,000,000$  meters

If production is  $x$  meters, then profit based on  $x$  is  $(x - 30,000,000) \times (275 - 225) = (x - 30,000,000) \times 50 = 6,000,000$ , thus  $x = 6,000,000 / 50 + 30,000,000 = 31,200,000$  meters. Hence production cost is  $31,200,000 \times 225 = 7,020,000,000 = 702$  Crores

Assume **demand** is 225 units per 10 minutes while **capacity per machine** is 150 units per 15 minutes, and there are FIVE (05) 5 machines of the same type and efficiency. Find the flow rate/throughput.

Range 1349 to 1351

Demand is  $225 \times 6 = 1350$  units per  $10 \times 6 = 60$  minutes (1 hour) while Capacity (ability to process) per machine is  $150 \times 4 = 600$  units per  $15 \times 4 = 60$  minutes (1 hour). Hence Capacity (ability to process) considering all the 5 machines =  $600 \times 5 = 3000$

Hence Flow Rate or Throughput =  $\min(225 \times 6, 150 \times 4 \times 5) = \min(1350, 3000) = 1350$  units per hour, and it is a Demand Constrained process

In a department store arrival rate is 450 customers per hour while allowed capacity rate is 100 customers per 1 hour per 1 billing desk. There are 5 billing desks, each of which have the same efficiency of working. The saving (in minutes considering 1 place of decimal) by the department store in an EIGHT (08) hour shift, using all FIVE (05) of the billing desks is?

Range 47.9 to 48.1

- Cycle Time =  $(60 \times 60) / 450 = 8.0$  seconds per customer.

- Capacity =  $(60 \times 60) / (5 \times 100) = 7.2$  seconds per customer.

Departmental store is able to handle the as it saves  $(8.0 - 7.2) = 0.8$  seconds per customer per hour, hence  $\{(0.8 \times 450) / (60 \times 60)\} \times 8$  hours = 0.8 hours in the eight-hour shift =  $0.8 \times 60 = 48$  minutes in the eight-hour shift

A primary hospital has the **capacity** to see 90 patients per 1 hour. The average **demand** rate is 320 patients per 4 hours. What is the utilization in percentage for the hospital (answer to 2 places of decimal)?

Range | 88.88 to 89.00

The calculations are shown below

- Capacity is  $4 \times 90 = 360$  patients per 4 hours

- Demand is 320 patients per 4 hours

- Flow Rate = Minimum {Demand, Process Capacity} =  $\text{Min}(320, 360) = 320$  patients per 4 hours

- Utilization = {Flow Rate/Capacity} =  $(320/360) = 0.8889$  which is 88.89%

For a production decision the outflow (in 10,000 INR), inflow (in 10,000 INR) and CCI (% pa) for the months of Jan to June are shown in the table below. Find the **Net Present Value** (NPV) in INR of the production decision considering you want to do that based at the **beginning** of Jan. Answer should be to two places of decimal.

| Month | Outflow (in 10,000 INR) at the end of month | Inflow (in 10,000 INR) at the end of month | CCI (% pa) |
|-------|---------------------------------------------|--------------------------------------------|------------|
| Jan   | 110                                         | 0                                          | 2.5        |
| Feb   | 120                                         | 80                                         | 2.4        |
| Mar   | 130                                         | 100                                        | 2.3        |
| Apr   | 140                                         | 130                                        | 2.2        |
| May   | 100                                         | 170                                        | 2.1        |
| Jun   | 50                                          | 180                                        | 2.0        |

Range | 87302 to 87304

| Month | Outflow (in 10,000 INR) at the end of month | Inflow (in 10,000 INR) at the end of month | CCI (% pa) | Time (Months) | NPV      |
|-------|---------------------------------------------|--------------------------------------------|------------|---------------|----------|
| Jan   | 110                                         | 0                                          | 2.5        | 0.083333333   | -1097711 |
| Feb   | 120                                         | 80                                         | 2.4        | 0.166666667   | -398403  |
| Mar   | 130                                         | 100                                        | 2.3        | 0.25          | -298280  |
| Apr   | 140                                         | 130                                        | 2.2        | 0.333333333   | -99269.3 |
| May   | 100                                         | 170                                        | 2.1        | 0.416666667   | 693901.7 |
| Jun   | 50                                          | 180                                        | 2          | 0.5           | 1287065  |
|       |                                             |                                            |            |               | 87303.29 |

For a production process the value of fixed cost ( $A$ ) is Rs. 1400/-, Demand ( $D$ ) is 7000 units per year,

variable cost ( $v$ ) per unit is Rs. 35/- and carrying cost ( $r$ ) is 25. What is the total cost (in INR) when order quantity is 15% less than the economic order quantity (EOQ), answer to one place of decimal?

Range 377690 to 377692

$$EOQ = Q^* = \sqrt{(2AD/vr)} = \sqrt{(2 \times 1400 \times 7000 / 35 \times 25)} = 149.663 \text{ units}$$

$$Q = 0.85 \times 149.663 = 127.2164$$

$$TRC(Q) = AD/Q + Dv + Qvr/2 = 1400 \times 7000 / 127.2164 + 7000 \times 35 + 127.2164 \times 35 \times 25 / 2 = 377691.3 \text{ in INR}$$

Given the usage of number of machines, Annual Fixed Cost (INR), Output (units), Variable/Unit (INR), Revenue/Unit (INR), then how many machine(s) it is plausible to operate, utilizing the idea of **Break Even Point (BEP)** is used.

**(There is one CORRECT answer out of the FIVE (05) options)**

| Number of Machines | Annual Fixed Cost (INR) | Output (Units) | Variable Cost/Unit (INR) | Revenue/Unit (INR) |
|--------------------|-------------------------|----------------|--------------------------|--------------------|
| 1                  | 190000                  | 0 to 500       | 110                      | 500                |
| 2                  | 250000                  | 501 to 1000    | 110                      | 500                |
| 3                  | 260000                  | 1001 to 1300   | 100                      | 510                |
| 4                  | 270000                  | 1301 to 1600   | 90                       | 530                |

A 1, 2 and 3

B 1 and 3

C 2 and 3

D 1 and 2

E 2 and 4

| Number of Machines | Annual Fixed Cost (INR) | Output (Units) | Variable Cost/Unit (INR) | Revenue/Unit (INR) | BEP      | Possibility |
|--------------------|-------------------------|----------------|--------------------------|--------------------|----------|-------------|
| 1                  | 190000                  | 0 to 500       | 110                      | 500                | 487.1795 | Yes         |
| 2                  | 250000                  | 501 to 1000    | 110                      | 500                | 641.0256 | Yes         |
| 3                  | 260000                  | 1001 to 1300   | 100                      | 510                | 634.1463 | No          |
| 4                  | 270000                  | 1301 to 1600   | 90                       | 530                | 613.6364 | No          |

Given the information below, what is the **difference** between sum of PV for Outflow and sum of PV for Inflow, in INR, where **Prevent Value (PV)** for BOTH are calculated at the beginning of Jan. Answer to 2 places of decimal, and consider the **numeric values** of PVs for **BOTH** Outflow and Inflow without the signs.

| IRR   | 5% pa                                  |                                       |
|-------|----------------------------------------|---------------------------------------|
| Month | Outflow (Lakh INR) at the end of month | Inflow (Lakh INR) at the end of month |
| Jan   | 220                                    | 190                                   |
| Mar   | 210                                    | 200                                   |

|              |                                                               |                                                              |                      |                                                              |                                                         |           |     |
|--------------|---------------------------------------------------------------|--------------------------------------------------------------|----------------------|--------------------------------------------------------------|---------------------------------------------------------|-----------|-----|
| <b>May</b>   |                                                               |                                                              |                      | 200                                                          |                                                         |           | 210 |
| <b>Jul</b>   |                                                               |                                                              |                      | 190                                                          |                                                         |           | 220 |
| Range        | -81958 to -81956                                              |                                                              |                      |                                                              |                                                         |           |     |
| <b>IRR</b>   | <b>5% pa</b>                                                  |                                                              |                      |                                                              |                                                         |           |     |
| <b>Month</b> | <b>Outflow<br/>(Lakh<br/>INR) at<br/>the end<br/>of month</b> | <b>Inflow<br/>(Lakh<br/>INR) at<br/>the end of<br/>month</b> | <b>Time<br/>(Yr)</b> | <b>PV Outflow<br/>(Lakh INR) at<br/>the end of<br/>month</b> | <b>PV Inflow<br/>(Lakh INR)<br/>at end of<br/>month</b> |           |     |
| Jan          | -220                                                          | 190                                                          | 0.083333             | -21908524.04                                                 | 18920998.04                                             |           |     |
| Mar          | -210                                                          | 200                                                          | 0.25                 | -20739133.81                                                 | 19751556.01                                             |           |     |
| May          | -200                                                          | 210                                                          | 0.416667             | -19587643.63                                                 | 20567025.81                                             |           |     |
| Jul          | -190                                                          | 220                                                          | 0.583333             | -18453836.93                                                 | 21367600.65                                             |           |     |
|              |                                                               |                                                              |                      | -80689138.41                                                 | 80607180.51                                             | -81957.90 |     |

|                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To derive the Economic Order Quantity (EOQ) the following assumptions are necessary<br>(There is one CORRECT answer out of the FIVE (05) options) |                                                                                                                                                                                                                                                                                                                                                                                              |
| A                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Demand rate is constant and deterministic</li> <li>• There are discounts for unit variable cost</li> <li>• Replenishment lead time is of zero</li> <li>• There is no inflation</li> <li>• Replenishment lead time is of zero</li> <li>• Entire order quantity is delivered at the same time</li> <li>• Planning horizon is very long</li> </ul>     |
| B                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Demand rate is constant and deterministic</li> <li>• There are no discounts for unit variable cost</li> <li>• Replenishment lead time is of zero</li> <li>• There is no inflation</li> <li>• Replenishment lead time is of zero</li> <li>• Entire order quantity is delivered in batches</li> <li>• Planning horizon is very long</li> </ul>        |
| C                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Demand rate is constant and deterministic</li> <li>• There are no discounts for unit variable cost</li> <li>• Replenishment lead time is of zero</li> <li>• There is no inflation</li> <li>• Replenishment lead time is non zero</li> <li>• Entire order quantity is delivered at the same time</li> <li>• Planning horizon is very long</li> </ul> |
| D                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Demand rate is constant and deterministic</li> </ul>                                                                                                                                                                                                                                                                                                |

|   |                                                                                                                                                                                                                                                                                                                                                                                             |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <ul style="list-style-type: none"> <li>• There are no discounts for unit variable cost</li> <li>• Replenishment lead time is of zero</li> <li>• There is inflation</li> <li>• Replenishment lead time is of zero</li> <li>• Entire order quantity is delivered at the same time</li> <li>• Planning horizon is very long</li> </ul>                                                         |
| E | <ul style="list-style-type: none"> <li>• Demand rate is constant and deterministic</li> <li>• There are no discounts for unit variable cost</li> <li>• Replenishment lead time is of zero</li> <li>• There is no inflation</li> <li>• Replenishment lead time is of zero</li> <li>• Entire order quantity is delivered at the same time</li> <li>• Planning horizon is very long</li> </ul> |

Order for slow moving/low requirement items is  $X \sim \text{NB}(0.3, 7)$ , while the demand for the same is  $X \sim \text{P}(17)$ . Given this information find the value of **Bull whip** effect?

(There is one **CORRECT** answer out of the **FIVE (05)** options)

|   |      |
|---|------|
| A | 0.88 |
| B | 1.79 |
| C | 2.34 |
| D | 3.10 |
| E | 3.20 |

Order for slow moving/low requirement items is  $X \sim \text{NB}(p=0.3, r=7)$ , hence  $\text{Variance} = r(1-p)/p^2 = 7 \times (1-0.3)/0.3^2 = 54.44$  and Demand for the same items is  $X \sim \text{P}(\lambda=17)$ , hence  $\text{Variance} = 17$ . Thus, Bullwhip value =  $54.44/17 = 3.20$

Given the pdf of **Gamma distribution**  $f(x) = \beta^{-\alpha} \Gamma^{-1}(\alpha)(x-\gamma)^{(\alpha-1)} \exp\{-(x-\gamma)/\beta\}$ ,  $x > \gamma$ , where  $\alpha > 0$ ,  $\beta > 0$  and  $\gamma$  are the shape, scale parameter and location parameters respectively then the expected value is  $\alpha/\beta - \gamma$

False

For the standard normal distribution, the points of inflexions are  $-1$  and  $+1$

True

SEVEN (07) equally competent quality control workers check SIXTY (60) numbers of 240 litre

fridges in a time span of FOUR (04) hours. It is also given that the value of output is Rs. 1,99,500/- while labour and electricity costs are Rs. 49,478/- and Rs. 33,621/- respectively. Then the ratio of multi-factor productivity wrt labour productivity is (units is hour/fridge) and answer to two places of decimal?

|       |              |
|-------|--------------|
| Range | 1.11 to 1.12 |
|-------|--------------|

Given the information we have

- Labour productivity =  $60 \text{ fridges} / (7 \text{ workers} \times 4 \text{ hours}) = 2.14 \text{ fridges/hour}$ .

- Multi-factor productivity =  $199500 / (49478 + 33621) = 2.40$

Hence: Multi-factor productivity/Labour productivity =  $(2.40/2.14) = 1.12 \text{ hour/fridge}$

During a typical Saturday evening, McDonalds at Z Square Mall in Kanpur serves 75 customers per hour when the eatery is open. A customer spends, on an average, 8 minutes in the shop. Thus on an average, how many customers are there in McDonalds simultaneously? Answer to one place of decimal?

|       |             |
|-------|-------------|
| Range | 9.9 to 10.1 |
|-------|-------------|

Inventory = Flow Rate  $\times$  Flow Time

Now Flow Rate =  $75/60 = 1.25 \text{ customer/minute}$  and Flow Time = 8 minutes

Inventory =  $1.25 \times 8 = 10 \text{ customers}$

There are 5 machines ( $i=1,2,3,4,5$ ) in parallel on the shop floor and all the 5 machines are doing the SAME operation. The average processing times for 3 machines ( $i=1,3,5$ ) are 4, 3 and 5 minutes for **EACH** job while for the other 2 machines ( $i=2,4$ ) the values are 12 and 10 minutes per TWO (02) jobs respectively. The average throughput for the shop floor per day (considering 24 hours) where for each day the shop floor operates for a TWO (02), EIGHT (08) hour shift is (answer to 0 place of decimal)?

**(There is one CORRECT answer out of the FIVE (05) options)**

|   |      |
|---|------|
| A | 1104 |
|---|------|

|   |      |
|---|------|
| B | 1140 |
|---|------|

|   |      |
|---|------|
| C | 1401 |
|---|------|

|   |      |
|---|------|
| D | 1410 |
|---|------|

|   |      |
|---|------|
| E | 1014 |
|---|------|

The calculations are shown below

- Machine ( $i=1$ ) on an average operates  $60/4 = 15$  jobs her hour

- Machine ( $i=2$ ) on an average operates  $60/(12/2) = 10$  jobs her hour

- Machine ( $i=3$ ) on an average operates  $60/3 = 20$  jobs her hour

- Machine ( $i=4$ ) on an average operates  $60/(10/2) = 12$  jobs her hour

- Machine ( $i=5$ ) on an average operates  $60/5 = 12$  jobs per hour  
Hence total  $8 \times 2 \times (15+10+20+12+12) = 8 \times 15 = 1104$  jobs per day

Assume Demand is 120 units per 30 minutes while Capacity per machine is 50 units per 15 minutes.  
Then what is the value of flow rate in units per hour?  
(There is one CORRECT answer out of the FIVE (05) options)

|   |     |
|---|-----|
| A | 120 |
| B | 240 |
| C | 200 |
| D | 100 |
| E | 180 |

Given the data that demand is 120 units per 30 minutes while Capacity per machine is 50 units per 15 minutes, we have

Demand =  $120 \times 2 = 240$  units per hour

Capacity =  $50 \times 4 = 200$  units per hour

Flow rate =  $\text{Min}(\text{Demand}, \text{Capacity}) = \text{Min}(240, 200) = 200$  units per hour

Consider a three-step airport security at Indira Gandhi International Airport (IGIA) (DEL), where

- First step ( $i=1$ ): Verifying ID and boarding pass and it takes THIRTY (30) seconds per passenger.
- Second step ( $i=2$ ): Security check and use of metal detector and it takes TWENTY (20) seconds per passenger.
- Third step ( $i=3$ ): Security check for hand luggage through an X-ray machine and it takes FIFTY FIVE (55) seconds per passenger bag.

If there are THIRTY (30) passengers standing in the queue and with **no other passengers in front of them**, then the total time taken (in hours) by the security system is, and answer to two places of decimal?  
(There is one CORRECT answer out of the FIVE (05) options)

|   |      |
|---|------|
| A | 0.37 |
| B | 0.47 |
| C | 0.52 |
| D | 0.57 |
| E | 0.58 |

Given the information, it takes  $(30 + 55)$  seconds = 85 seconds until the FIRST customer is served.

From then onward, we are serving a customer every 55 seconds, and as we have 30 customers to serve, this will take  $(30 - 1) \times 55$  seconds =  $29 \times 55 = 1595$  seconds

Total time taken  $(85 + 1595)$  seconds =  $1680/60 = 28$  minutes =  $28/60 = 0.47$  hour

Consider a FIVE (05) stage production system, where

- **First stage ( $i=1$ ):** ONE (01) unit of **item # 01** is processed in THREE (03) minutes.
- **Second stage ( $i=2$ ):** ONE (01) unit of **item # 02** is processed in TWO (02) minutes.
- **Third stage ( $i=3$ ):** ONE (01) unit of **item # 03** is processed in SIX (06) minutes, whereby ONE (01) unit of **item # 03** requires: ONE (01) unit of **item # 01** (obtained from First stage ( $i=1$ )) and TWO (02) units of **item # 02** (obtained from Second stage ( $i=2$ )).
- **Fourth stage ( $i=4$ ):** ONE (01) unit of **item # 04** is processed in FOUR (04) minutes.
- **Fifth stage ( $i=5$ ):** ONE (01) unit of **item # 05** is processed in ONE (01) minute, whereby ONE (01) unit of **item # 05** requires: TWO (02) unit of **item # 03** (obtained from Third stage ( $i=3$ )) and THREE (03) units of **item # 04** (obtained from Fourth stage ( $i=4$ )).

Find the **total time** (in minutes) to produce a batch of TEN (10) units of **item # 05**?

(There is one CORRECT answer out of the FIVE (05) options)

|   |     |
|---|-----|
| A | 197 |
| B | 198 |
| C | 199 |
| D | 201 |
| E | 203 |

To produce ONE (01) unit of Item # 05 we required  $1 \times 20 + 01 = 21$  minutes, and the general formulae is  $i \times 20 + 01 = 21$  here  $i = \text{Number of Item \# 05 produced}$ , hence to produce TEN (10) units of Item # 05 we required  $10 \times 20 + 01 = 201$  minutes, here  $i = 10$

We can draw the GANTT chart to visualize the problem and it is as follows

| Time      | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |
|-----------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|
| Item # 01 | 1 | 1 | 1 |   |   |   |   |   |   |    | 2  | 2  | 2  |    |    |    |    |    |    |    |    |
| Item # 02 | 1 | 1 | 2 | 2 |   |   |   |   |   |    | 3  | 3  | 4  | 4  |    |    |    |    |    |    |    |
| Item # 03 |   |   |   |   | 1 | 1 | 1 | 1 | 1 | 1  |    |    |    |    | 2  | 2  | 2  | 2  | 2  | 2  |    |
| Item # 04 |   |   |   |   |   |   |   |   | 1 | 1  | 1  | 1  | 2  | 2  | 2  | 2  | 3  | 3  | 3  | 3  |    |
| Item # 05 |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    | 1  |

GANTT charts are used as?

(There is one CORRECT answer out of the FIVE (05) options)

|   |                                                                                        |
|---|----------------------------------------------------------------------------------------|
| A | Project management tool to plan, schedule, and find the maximum deterministic time     |
| B | Project management tool to plan, schedule, and find the minimum deterministic time     |
| C | Project management tool to plan, schedule, and find the maximum permissible stochastic |

|                                                                                   |                                                                                             |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|                                                                                   | time                                                                                        |
| D                                                                                 | Project management tool to plan, schedule, and find the minimum permissible stochastic time |
| E                                                                                 | Project management tool to plan, schedule, and track project with respect of time           |
| Project management tool to plan, schedule, and track project with respect of time |                                                                                             |

Given the following information

- $Q$  = Replenishment order quantity (in **units**)
- $D$  = Demand rate of the item (in **units per year**)
- $T = Q/D$  = Time between placement (and receipt) of replenishment orders (in **years**)
- $N = D/Q$  = Number of replenishment **number per year**
- $A$  = Fixed cost component with each replenishment (independent of magnitude of replenishment) (in **Rs.**)
- $v$  = Variable cost per item (in **Rs. per unit**)
- $r$  = Carrying cost/cost of having Re. 1/- of the item tied up in the inventory for a unit time interval (normally 1 year) (in **Rs. per Rs. per unit time**)

The formulae for **Total Relevant Cost** (wrt to  $T^*$ ) is given as

**(There is one CORRECT answer out of the FIVE (05) options)**

|                             |                              |
|-----------------------------|------------------------------|
| A                           | $(A/T^*) + (DT^*vr)/2 + D/v$ |
| B                           | $(A/T^*) + (D/T^*)vr/2 + Dv$ |
| C                           | $(A/T^*) + (DT^*vr)/2 + Dv$  |
| D                           | $(A/T^*) + (DT^*v)/2r + Dv$  |
| E                           | $(A/T^*) + (DT^*r)/2v + D/v$ |
| $(A/T^*) + (DT^*vr)/2 + Dv$ |                              |

Considering the process life cycle stages the production process based on the number of products produced can be ranked (from MAXIMUM number of production to MINIMUM number of production) as?

**(There is one CORRECT answer out of the FIVE (05) options)**

|   |                                                                                                   |
|---|---------------------------------------------------------------------------------------------------|
| A | Continuous Flow Production<br>Job Shop Production<br>Batch Production<br>Assembly Line Production |
| B | Job Shop Production<br>Batch Production<br>Assembly Line Production                               |

|   |                                                                                                   |
|---|---------------------------------------------------------------------------------------------------|
|   | Continuous Flow Production                                                                        |
| C | Continuous Flow Production<br>Assembly Line Production<br>Batch Production<br>Job Shop Production |
| D | Continuous Flow Production<br>Batch Production<br>Assembly Line Production<br>Job Shop Production |
| E | Continuous Flow Production<br>Batch Production<br>Job Shop Production<br>Assembly Line Production |
|   | Continuous Flow Production<br>Assembly Line Production<br>Batch Production<br>Job Shop Production |